



# **GCSE MARKING SCHEME**

**SUMMER 2023**

**GCSE  
CHEMISTRY – UNIT 2**

**3410U20-1 AND 3410UB0-1**

## **INTRODUCTION**

This marking scheme was used by WJEC for the 2023 examination. It was finalised after detailed discussion at examiners' conferences by all the examiners involved in the assessment. The conference was held shortly after the paper was taken so that reference could be made to the full range of candidates' responses, with photocopied scripts forming the basis of discussion. The aim of the conference was to ensure that the marking scheme was interpreted and applied in the same way by all examiners.

It is hoped that this information will be of assistance to centres but it is recognised at the same time that, without the benefit of participation in the examiners' conference, teachers may have different views on certain matters of detail or interpretation.

WJEC regrets that it cannot enter into any discussion or correspondence about this marking scheme.

## GCSE CHEMISTRY UNIT 2

### CHEMICAL BONDING, APPLICATION OF CHEMICAL REACTIONS AND ORGANIC CHEMISTRY

#### SUMMER 2023 MARK SCHEME

#### GENERAL INSTRUCTIONS

##### Marking rules

All work should be seen to have been marked.

Marking schemes will indicate when explicit working is deemed to be a necessary part of a correct answer.

Crossed out responses not replaced should be marked.

Credit will be given for correct and relevant alternative responses which are not recorded in the mark scheme.

##### Extended response question

A level of response mark scheme is used. Before applying the mark scheme please read through the whole answer from start to finish. Firstly, decide which level descriptor matches best with the candidate's response: remember that you should be considering the overall quality of the response. Then decide which mark to award within the level. Award the higher mark in the level if there is a good match with both the content statements and the communication statements.

##### Marking abbreviations

The following may be used in marking schemes or in the marking of scripts to indicate reasons for the marks awarded.

|     |   |                       |
|-----|---|-----------------------|
| cao | = | correct answer only   |
| ecf | = | error carried forward |
| bod | = | benefit of doubt      |

## FOUNDATION TIER ONLY QUESTIONS

| Question |     |       |  | Marking details                                                                                                                           | Marks available |     |     |       |       |      |
|----------|-----|-------|--|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                           | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 1        | (a) | (i)   |  | accept <b>either</b> of following <ul style="list-style-type: none"> <li>hand wash</li> <li>drain cleaner</li> </ul> accept blue / purple |                 |     | 1   | 1     |       | 1    |
|          |     | (ii)  |  | battery fluid<br>accept red                                                                                                               |                 |     | 1   | 1     |       | 1    |
|          |     | (iii) |  | NaOH                                                                                                                                      |                 | 1   |     | 1     |       |      |
|          | (b) | (i)   |  | <b>C</b> (1)<br>accept sodium carbonate<br>fizzing (1)                                                                                    |                 |     | 2   | 2     |       | 2    |
|          |     | (ii)  |  | lighted / burning splint (1)<br>do <b>not</b> accept <u>glowing</u> splint<br>gives squeaky pop (1)                                       | 2               |     |     | 2     |       | 2    |

| Question |  |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                            | Marks available |          |          |          |          |          |
|----------|--|-------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                            | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
|          |  | (iii) |  | <b>C (1)</b><br>accept sodium carbonate<br><br>award (1) for <b>any</b> of following <ul style="list-style-type: none"> <li>• lowest (temperature) increase</li> <li>• (temperature) increases less</li> <li>• (temperature) increases only a little</li> <li>• temperature <u>only</u> goes up by 5°C</li> </ul> accept least energy given out<br><br>neutral answer – lowest temperature |                 |          | 2        | 2        |          | 2        |
|          |  |       |  | <b>Question 1 total</b>                                                                                                                                                                                                                                                                                                                                                                    | <b>2</b>        | <b>1</b> | <b>6</b> | <b>9</b> | <b>0</b> | <b>8</b> |

| Question                                      |     |      |  | Marking details                                                                                                                                                                                                                                                                                      | Marks available                               |     |                                              |       |                                             |      |                                      |  |  |  |   |   |  |   |
|-----------------------------------------------|-----|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-----|----------------------------------------------|-------|---------------------------------------------|------|--------------------------------------|--|--|--|---|---|--|---|
|                                               |     |      |  |                                                                                                                                                                                                                                                                                                      | AO1                                           | AO2 | AO3                                          | Total | Maths                                       | Prac |                                      |  |  |  |   |   |  |   |
| 2                                             | (a) | (i)  |  | heat<br><br>accept ignition                                                                                                                                                                                                                                                                          | 1                                             |     |                                              | 1     |                                             |      |                                      |  |  |  |   |   |  |   |
|                                               |     | (ii) |  | oxygen<br><br>accept air                                                                                                                                                                                                                                                                             | 1                                             |     |                                              | 1     |                                             |      |                                      |  |  |  |   |   |  |   |
|                                               |     |      |  | fuel                                                                                                                                                                                                                                                                                                 | 1                                             |     |                                              | 1     |                                             |      |                                      |  |  |  |   |   |  |   |
|                                               | (b) |      |  | 4                                                                                                                                                                                                                                                                                                    |                                               | 1   |                                              | 1     | 1                                           |      |                                      |  |  |  |   |   |  |   |
|                                               | (c) | (i)  |  | 37                                                                                                                                                                                                                                                                                                   |                                               | 1   |                                              | 1     | 1                                           | 1    |                                      |  |  |  |   |   |  |   |
|                                               |     | (ii) |  | <table><tr><td>Which alcohol gives out the most heat energy?</td><td>✓</td></tr><tr><td>Which gases are produced when alcohols burn?</td><td></td></tr><tr><td>Which alcohol has the lowest boiling point?</td><td></td></tr><tr><td>Which alcohol burns for the longest?</td><td></td></tr></table> | Which alcohol gives out the most heat energy? | ✓   | Which gases are produced when alcohols burn? |       | Which alcohol has the lowest boiling point? |      | Which alcohol burns for the longest? |  |  |  | 1 | 1 |  | 1 |
| Which alcohol gives out the most heat energy? | ✓   |      |  |                                                                                                                                                                                                                                                                                                      |                                               |     |                                              |       |                                             |      |                                      |  |  |  |   |   |  |   |
| Which gases are produced when alcohols burn?  |     |      |  |                                                                                                                                                                                                                                                                                                      |                                               |     |                                              |       |                                             |      |                                      |  |  |  |   |   |  |   |
| Which alcohol has the lowest boiling point?   |     |      |  |                                                                                                                                                                                                                                                                                                      |                                               |     |                                              |       |                                             |      |                                      |  |  |  |   |   |  |   |
| Which alcohol burns for the longest?          |     |      |  |                                                                                                                                                                                                                                                                                                      |                                               |     |                                              |       |                                             |      |                                      |  |  |  |   |   |  |   |
|                                               | (d) |      |  | 32 (2)<br><br>if answer incorrect award (1) for any clear indication of the correct number of all atoms e.g.<br><br>• 12 + 4 + 16<br>• C + 4H + O<br><br>no ecf possible                                                                                                                             |                                               | 2   |                                              | 2     | 2                                           |      |                                      |  |  |  |   |   |  |   |
|                                               |     |      |  | Question 2 total                                                                                                                                                                                                                                                                                     | 3                                             | 4   | 1                                            | 8     | 4                                           | 2    |                                      |  |  |  |   |   |  |   |

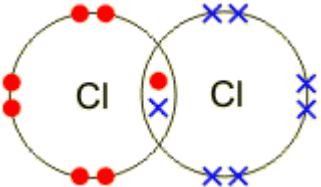
| Question                               |     |       |  | Marking details                                                                                                                                                                                | Marks available          |     |                      |       |                                        |      |   |  |  |   |  |   |
|----------------------------------------|-----|-------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-----|----------------------|-------|----------------------------------------|------|---|--|--|---|--|---|
|                                        |     |       |  |                                                                                                                                                                                                | AO1                      | AO2 | AO3                  | Total | Maths                                  | Prac |   |  |  |   |  |   |
| 3                                      | (a) | (i)   |  | conical flask                                                                                                                                                                                  | 1                        |     |                      | 1     |                                        | 1    |   |  |  |   |  |   |
|                                        |     | (ii)  |  | limewater (1)<br>goes milky / cloudy (1)                                                                                                                                                       | 2                        |     |                      | 2     |                                        | 2    |   |  |  |   |  |   |
|                                        |     | (iii) |  | <table border="1"><tr><td>the reaction is finished</td><td></td></tr><tr><td>the yeast is used up</td><td></td></tr><tr><td>the enzymes in the yeast are denatured</td><td>✓</td></tr></table> | the reaction is finished |     | the yeast is used up |       | the enzymes in the yeast are denatured | ✓    | 1 |  |  | 1 |  | 1 |
| the reaction is finished               |     |       |  |                                                                                                                                                                                                |                          |     |                      |       |                                        |      |   |  |  |   |  |   |
| the yeast is used up                   |     |       |  |                                                                                                                                                                                                |                          |     |                      |       |                                        |      |   |  |  |   |  |   |
| the enzymes in the yeast are denatured | ✓   |       |  |                                                                                                                                                                                                |                          |     |                      |       |                                        |      |   |  |  |   |  |   |
|                                        | (b) | (i)   |  | award (2) for <b>all</b> points plotted correctly<br>tolerance $\pm\frac{1}{2}$ square<br><br>award (1) for <b>any</b> 4 correct<br><br>suitable straight line drawn (with ruler) (1)          |                          | 3   |                      | 3     | 3                                      |      |   |  |  |   |  |   |
|                                        |     | (ii)  |  | 15<br><br>accept any value in the range 14-16<br><br>ecf possible from incorrect graph                                                                                                         |                          | 1   |                      | 1     | 1                                      |      |   |  |  |   |  |   |
|                                        |     | (iii) |  | accept any value in the range 160-172<br><br>no ecf possible                                                                                                                                   |                          | 1   |                      | 1     | 1                                      |      |   |  |  |   |  |   |
|                                        |     |       |  | Question 3 total                                                                                                                                                                               | 4                        | 5   | 0                    | 9     | 5                                      | 4    |   |  |  |   |  |   |

| Question                               |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks available                        |     |                                      |       |                                        |      |   |  |  |   |  |  |
|----------------------------------------|-----|-------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-----|--------------------------------------|-------|----------------------------------------|------|---|--|--|---|--|--|
|                                        |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | AO1                                    | AO2 | AO3                                  | Total | Maths                                  | Prac |   |  |  |   |  |  |
| 4                                      | (a) | (i)   |  | compound (1)<br><br>lower (1)<br><br>electrical (1)<br><br>liquid (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4                                      |     |                                      | 4     |                                        |      |   |  |  |   |  |  |
|                                        |     | (ii)  |  | Al <sub>2</sub> O <sub>3</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                        | 1   |                                      | 1     |                                        |      |   |  |  |   |  |  |
|                                        |     | (iii) |  | <table border="1"><tr><td>carbon is more reactive than aluminium</td><td></td></tr><tr><td>iron is more reactive than aluminium</td><td></td></tr><tr><td>aluminium is more reactive than carbon</td><td>✓</td></tr></table>                                                                                                                                                                                                                                                                                                                      | carbon is more reactive than aluminium |     | iron is more reactive than aluminium |       | aluminium is more reactive than carbon | ✓    | 1 |  |  | 1 |  |  |
| carbon is more reactive than aluminium |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                        |     |                                      |       |                                        |      |   |  |  |   |  |  |
| iron is more reactive than aluminium   |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                        |     |                                      |       |                                        |      |   |  |  |   |  |  |
| aluminium is more reactive than carbon | ✓   |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                        |     |                                      |       |                                        |      |   |  |  |   |  |  |
|                                        | (b) |       |  | <b>Indicative content</b><br><br><b>electrical wiring</b> <ul style="list-style-type: none"><li>ductile – can be drawn into wires</li><li>good electrical conductor – current can pass through it</li></ul><br><b>saucepans</b> <ul style="list-style-type: none"><li>good thermal conductor – heat can pass through it</li><li>high melting point – can be heated to high temperatures</li><li>corrosion resistant – will not corrode</li><li>malleable – can be hammered into shape / shaped</li><li>non-toxic – safe to use for food</li></ul> | 4                                      |     | 2                                    | 6     |                                        |      |   |  |  |   |  |  |

| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks available |     |     |       |       |      |
|----------|--|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |  |  |  | <p><b>water pipes</b></p> <ul style="list-style-type: none"><li>• malleable – can be hammered into shape / shaped</li><li>• corrosion resistant – will not corrode</li><li>• non-toxic – safe for (drinking) water</li></ul> <p>references to strength, durability, low/high density are not directly relevant to these uses</p> <p><b>5-6 marks</b><br/>Description of two relevant properties linked to all three uses<br/><i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i></p> <p><b>3-4 marks</b><br/>Description of one relevant property linked to all three uses<br/><i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i></p> <p><b>1-2 marks</b><br/>Identification of one relevant property linked to one or two uses<br/><i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i></p> <p><b>0 marks</b><br/><i>No attempt made or no response worthy of credit.</i></p> |                 |     |     |       |       |      |
|          |  |  |  | Question 4 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 9               | 1   | 2   | 12    | 0     | 0    |

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                        | Marks available |     |     |       |       |      |
|----------|-----|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                        | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 5        | (a) | (i)  |  | crop growth on fields increases                                                                                                                                                                                                                                                                                                                                                                        |                 |     |     |       |       |      |
|          |     |      |  | fertilisers run into waterways                                                                                                                                                                                                                                                                                                                                                                         |                 |     |     |       |       |      |
|          |     |      |  | plant growth in rivers and lakes increases                                                                                                                                                                                                                                                                                                                                                             |                 |     |     |       |       |      |
|          |     |      |  | aquatic animals do not have enough oxygen                                                                                                                                                                                                                                                                                                                                                              |                 |     |     |       |       |      |
|          |     |      |  | farmers' profits increase                                                                                                                                                                                                                                                                                                                                                                              |                 |     |     |       |       |      |
|          |     |      |  | award (2) for <b>all</b> 3 correct<br>award (1) for <b>any</b> 2 correct                                                                                                                                                                                                                                                                                                                               |                 |     | 2   | 2     |       |      |
|          |     | (ii) |  | award (1) for statement that in general the larger the population, the more ammonia is used e.g. Asia has the largest population and the largest ammonia use / the largest population uses the most ammonia<br>'demand for food' is equivalent to population<br><br>award (1) for recognising Africa as the exception e.g. Africa has the second largest population but the fourth largest ammonia use |                 |     | 2   | 2     | 2     |      |
|          | (b) | (i)  |  | ammonium sulfate<br><br>do not accept ammonia sulfate                                                                                                                                                                                                                                                                                                                                                  |                 | 1   |     | 1     |       |      |
|          |     | (ii) |  | plants use nitrogen to make sugar                                                                                                                                                                                                                                                                                                                                                                      |                 |     |     |       |       |      |
|          |     |      |  | plants use nitrogen to make water                                                                                                                                                                                                                                                                                                                                                                      |                 |     |     |       |       |      |
|          |     |      |  | plants use nitrogen to make oxygen                                                                                                                                                                                                                                                                                                                                                                     |                 |     |     |       |       |      |
|          |     |      |  | plants use nitrogen to make protein                                                                                                                                                                                                                                                                                                                                                                    |                 |     |     |       |       |      |
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                        | 1               |     |     | 1     |       |      |

| Question |     |       |    | Marking details                                                                                                                                                                                                                                                                                                         | Marks available |          |          |           |          |          |
|----------|-----|-------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |       |    |                                                                                                                                                                                                                                                                                                                         | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
|          |     | (iii) |    | 300 (2)<br><br>if answer incorrect award (1) for $\frac{75}{15} / 5 / 0.3$                                                                                                                                                                                                                                              |                 | 2        |          | 2         | 2        |          |
|          | (c) | (i)   |    | $\text{N}_2 + \boxed{3} \text{H}_2 \rightleftharpoons \boxed{2} \text{NH}_3$                                                                                                                                                                                                                                            |                 | 1        |          | 1         |          |          |
|          |     | (ii)  |    | award (1) for any of following <ul style="list-style-type: none"> <li>the reaction is reversible</li> <li>the reaction goes in both directions</li> <li>the reaction can go forwards or backwards</li> </ul> do not accept equal both ways                                                                              | 1               |          |          | 1         |          |          |
|          |     | (iii) | I  | iron / Fe                                                                                                                                                                                                                                                                                                               | 1               |          |          | 1         |          |          |
|          |     |       | II | award (1) for any of following <ul style="list-style-type: none"> <li>increases rate of reaction</li> <li>makes the reaction go faster / it go faster</li> <li>makes the ammonia more quickly</li> <li>makes the product more quickly</li> <li>process takes less time</li> </ul> accept 'lowers the activation energy' | 1               |          |          | 1         |          |          |
|          |     |       |    | <b>Question 5 total</b>                                                                                                                                                                                                                                                                                                 | <b>4</b>        | <b>4</b> | <b>4</b> | <b>12</b> | <b>4</b> | <b>0</b> |

| Question |     |       |  | Marking details                                                                                                                                                                                 | Marks available |     |     |       |       |      |
|----------|-----|-------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                 | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 6        | (a) | (i)   |  | <b>D and E</b> both needed                                                                                                                                                                      |                 |     | 1   | 1     |       |      |
|          |     | (ii)  |  | <b>B</b>                                                                                                                                                                                        |                 |     | 1   | 1     |       |      |
|          |     | (iii) |  | <b>C</b>                                                                                                                                                                                        |                 |     | 1   | 1     |       |      |
|          | (b) |       |  |  <p>shared pair (1)</p> <p>octet (1)</p> <p>accept with all dots/crosses<br/>ignore any inner shells drawn</p> |                 | 2   |     | 2     |       |      |
|          | (c) | (i)   |  | <p><b>V</b> 2+ (1) accept +2</p> <p><b>Y</b> 1- / - (1) accept -1</p> <p>award (1) for <b>V</b> ⇒ positive <b>and</b> <b>Y</b> ⇒ negative</p>                                                   |                 | 2   |     | 2     |       |      |
|          |     | (ii)  |  | <b>VY<sub>2</sub></b>                                                                                                                                                                           |                 | 1   |     | 1     |       |      |

| Question |  |       |  | Marking details                                                                                                                                                                                                                                                               | Marks available |     |     |       |       |      |
|----------|--|-------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |       |  |                                                                                                                                                                                                                                                                               | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |  | (iii) |  | <p>83 (2)</p> <p>award (1) for correct answer not to nearest whole number<br/>e.g. 82.9 / 82.98 / 82.97</p> <p>if answer incorrect award (1) for any of following<br/> <math>39 \times 2</math><br/>           78<br/>           82<br/>           0.83<br/>           41</p> |                 | 2   |     | 2     | 2     |      |
|          |  |       |  | Question 6 total                                                                                                                                                                                                                                                              | 0               | 7   | 3   | 10    | 2     | 0    |

## COMMON QUESTIONS

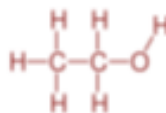
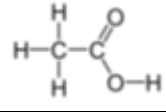
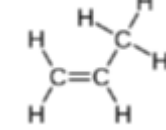
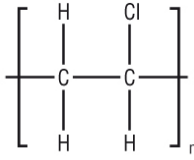
| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                           | Marks available |     |     |       |       |      |
|----------|-----|-------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                           | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 7/1      | (a) | (i)   |  | award (1) for any of following <ul style="list-style-type: none"> <li>• leave to crystallise / evaporate / dry naturally</li> <li>• leave to dry for a few days / until next lesson</li> <li>• leave to dry in a warm place / on window sill / on radiator</li> </ul> must have a 'process' and the idea that it happens over a period of time <b>OR</b> in a warm place<br>neutral answer – leave to dry | 1               |     |     | 1     |       | 1    |
|          |     | (ii)  |  | no fizzing / bubbles / effervescence (with oxide) (1)<br>because no carbon dioxide produced (1)<br>alternative answer<br>black powder (rather than green) would be left in the beaker when all the acid has reacted (1)<br>because copper(II) oxide is black (not green) (1)                                                                                                                              |                 |     | 2   | 2     |       | 2    |
|          |     | (iii) |  | $\text{CuSO}_4 + \text{H}_2\text{O}$<br>award (1) for each correct product                                                                                                                                                                                                                                                                                                                                |                 | 2   |     | 2     |       |      |

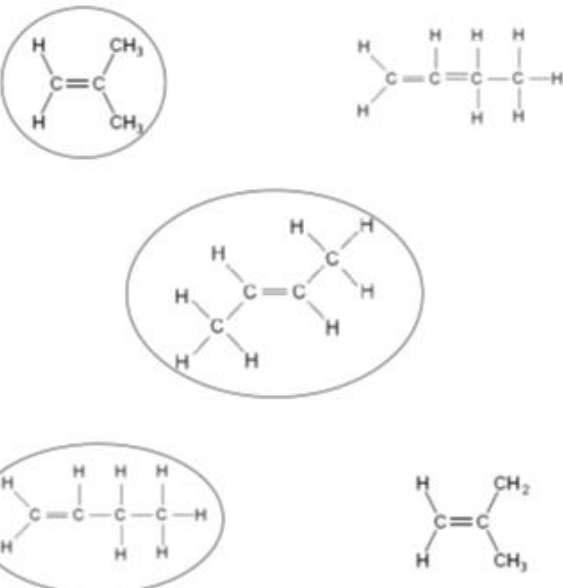
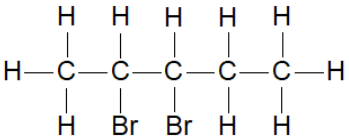
| Question                          |        |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                           |                            | Marks available |                                |     |                         |       |                                   |   |   |  |  |   |  |  |
|-----------------------------------|--------|-------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------------|--------------------------------|-----|-------------------------|-------|-----------------------------------|---|---|--|--|---|--|--|
|                                   |        |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                            | AO1             | AO2                            | AO3 | Total                   | Maths | Prac                              |   |   |  |  |   |  |  |
|                                   | (b)    | (i)   |  | <table><tr><td>Part of the energy profile</td><td>Letter</td></tr><tr><td>energy change for the reaction</td><td>C</td></tr><tr><td>energy of the reactants</td><td>A</td></tr><tr><td>activation energy of the reaction</td><td>B</td></tr></table><br>award (2) for <b>all</b> three correct<br>award (1) for <b>any</b> one correct                                                                                                    | Part of the energy profile | Letter          | energy change for the reaction | C   | energy of the reactants | A     | activation energy of the reaction | B | 2 |  |  | 2 |  |  |
| Part of the energy profile        | Letter |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                            |                 |                                |     |                         |       |                                   |   |   |  |  |   |  |  |
| energy change for the reaction    | C      |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                            |                 |                                |     |                         |       |                                   |   |   |  |  |   |  |  |
| energy of the reactants           | A      |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                            |                 |                                |     |                         |       |                                   |   |   |  |  |   |  |  |
| activation energy of the reaction | B      |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                            |                 |                                |     |                         |       |                                   |   |   |  |  |   |  |  |
|                                   |        | (ii)  |  | the (minimum) energy required for a reaction to happen / start<br><br>accept 'the <u>minimum</u> energy required to activate the reaction'<br>neutral answer – the energy required to activate the reaction                                                                                                                                                                                                                               | 1                          |                 |                                | 1   |                         |       |                                   |   |   |  |  |   |  |  |
|                                   |        | (iii) |  | award (1) for any of following<br><br><ul style="list-style-type: none"><li>the energy of the products is lower than the energy of the reactants</li><li>the product line is below the reactant line / <b>E</b> is below <b>A</b></li><li>energy given out is greater than energy taken in / <b>D</b> is greater than <b>B</b></li><li>lower energy at the end than at the beginning</li></ul><br>neutral answer – negative energy change |                            | 1               |                                | 1   |                         |       |                                   |   |   |  |  |   |  |  |
|                                   |        |       |  | Question 7/1 total                                                                                                                                                                                                                                                                                                                                                                                                                        | 4                          | 3               | 2                              | 9   | 0                       | 3     |                                   |   |   |  |  |   |  |  |

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                            | Marks available |     |     |       |       |      |
|----------|-----|-------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                            | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 8/2      | (a) | (i)   |  | <b>X</b> boiling / evaporation<br><b>Y</b> condensing / condensation <b>both</b> needed<br>neutral answers – liquid to gas / gas to liquid                                                                                                                                                 | 1               |     |     | 1     |       | 1    |
|          |     | (ii)  |  | award (2) for statement linking boiling point and chain length<br>e.g. the longer the chain length, the higher the boiling point<br><br>award (1) for either of following<br>different boiling points<br>different chain lengths<br><br>chain lengths and size of molecules are equivalent | 2               |     |     | 2     |       |      |
|          |     | (iii) |  | award (3) for 8300<br><br>award (2) for 8274 - answer not to two sig figs<br><br>if answer incorrect award (1) for temperature rise of 19.7<br><br>ecf possible from incorrect temperature rise                                                                                            |                 | 3   |     | 3     | 3     | 3    |
|          | (b) | (i)   |  | $C_6H_{12}$<br>accept $2C_3H_6$                                                                                                                                                                                                                                                            |                 | 1   |     | 1     |       |      |
|          |     | (ii)  |  | award (1) for any two conditions <ul style="list-style-type: none"> <li>high temperature / heat</li> <li>catalyst</li> <li>high pressure</li> </ul> accept 'high temperature and pressure'                                                                                                 | 1               |     |     | 1     |       |      |

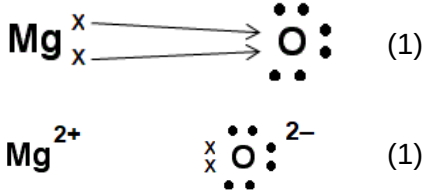
| Question |  |       |  | Marking details                                                                                                                                                                                                                                                                                                                             | Marks available |          |          |           |          |          |
|----------|--|-------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |  |       |  |                                                                                                                                                                                                                                                                                                                                             | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
|          |  | (iii) |  | award (1) for any of following <ul style="list-style-type: none"> <li>• has double bond between two carbon atoms</li> <li>• has C=C bond</li> <li>• has carbon atoms which could bond with more hydrogen / more atoms</li> <li>• could undergo an addition reaction</li> </ul> neutral answer – not completely surrounded by hydrogen atoms | 1               |          |          | 1         |          |          |
|          |  | (iv)  |  | $C_8H_{18}$ (one of the compounds) present in petrol / good fuel (1)<br>neutral answer – used in cars<br><br>$C_2H_4$ used to make plastics / polymers / polythene / ethanol (1)<br>neutral answer – fuel                                                                                                                                   | 2               |          |          | 2         |          |          |
|          |  |       |  | <b>Question 8/2 total</b>                                                                                                                                                                                                                                                                                                                   | <b>7</b>        | <b>4</b> | <b>0</b> | <b>11</b> | <b>3</b> | <b>4</b> |

## HIGHER TIER ONLY QUESTIONS

| Question |     |      |  | Marking details                                                                                                                                |                                  |                                                                                    |                   | Marks available |     |     |       |       |      |
|----------|-----|------|--|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|------------------------------------------------------------------------------------|-------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                |                                  |                                                                                    |                   | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 3        | (a) | (i)  |  | Name                                                                                                                                           | Molecular formula                | Structure                                                                          | Homologous series | 3               |     |     | 3     |       |      |
|          |     |      |  | ethanol                                                                                                                                        | C <sub>2</sub> H <sub>5</sub> OH |  | alcohols          |                 |     |     |       |       |      |
|          |     |      |  | ethanoic acid                                                                                                                                  | CH <sub>3</sub> COOH             |  | carboxylic acids  |                 |     |     |       |       |      |
|          |     |      |  | propene                                                                                                                                        | C <sub>3</sub> H <sub>6</sub>    |  | alkenes           |                 |     |     |       |       |      |
|          |     |      |  | award (1) for each correct answer                                                                                                              |                                  |                                                                                    |                   |                 |     |     |       |       |      |
|          |     | (ii) |  | C <sub>7</sub> H <sub>14</sub>                                                                                                                 |                                  |                                                                                    |                   |                 | 1   |     | 1     | 1     |      |
|          | (b) |      |  |  repeat unit (1)<br><br>n and side bonds (and brackets) (1) |                                  |                                                                                    |                   | 2               |     |     | 2     |       |      |

| Question |     |  |  | Marking details                                                                                                                         | Marks available |     |     |       |       |      |
|----------|-----|--|--|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                         | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (c) |  |  |  <p>all three correct (2)<br/>any two correct (1)</p> | 2               |     |     | 2     |       |      |
|          | (d) |  |  |                                                      |                 | 1   |     | 1     |       |      |
|          |     |  |  | Question 3 total                                                                                                                        | 7               | 2   | 0   | 9     | 1     | 0    |

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks available |          |          |          |          |          |
|----------|-----|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 4        | (a) |      |  | sodium<br>zinc<br>iron<br>lead                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                 |          | 1        | 1        |          | 1        |
|          | (b) | (i)  |  | award (2) for <b>all</b> points plotted correctly<br>tolerance $\pm \frac{1}{2}$ square<br>award (1) for <b>any</b> 4 correct (1)<br>suitable straight line drawn (with ruler) (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                 | 3        |          | 3        | 3        |          |
|          |     | (ii) |  | award (1) for either of following <ul style="list-style-type: none"> <li>order of reactivity is <math>\text{Mg} &gt; \text{Al} &gt; \text{Zn} &gt; \text{Cu}</math></li> <li>magnesium and aluminium are more reactive than zinc and copper is less reactive than zinc – must refer to all four metals</li> </ul> award (1) for any of following <ul style="list-style-type: none"> <li>copper does not react with zinc chloride / does not displace zinc</li> <li>temperature doesn't change/increase with copper</li> <li>aluminium and magnesium react with zinc chloride / displace zinc</li> </ul> award (1) for any of following <ul style="list-style-type: none"> <li>reaction between magnesium and zinc chloride is more exothermic than that between aluminium and zinc chloride</li> <li>magnesium reaction more exothermic than aluminium</li> <li>temperature increases more with magnesium than aluminium</li> <li>magnesium most exothermic</li> </ul> |                 |          | 3        | 3        |          | 3        |
|          |     |      |  | <b>Question 4 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>0</b>        | <b>3</b> | <b>4</b> | <b>7</b> | <b>3</b> | <b>4</b> |

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                              | Marks available |     |     |       |       |      |
|----------|-----|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                              | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 5        | (a) | (i)  |  | <br>four shared pairs (1)<br>both octets (1)                                                                                                                                                                                                                                |                 | 2   |     | 2     |       |      |
|          |     | (ii) |  | <u>intermolecular</u> forces are weak / forces <u>between</u> molecules are weak (1)<br>accept bonds / interactions / attractions<br>require little energy to overcome / break forces (1)<br>accept doesn't take much heat to overcome / break forces<br>neutral answer - simple molecular<br>do not award any credit if explanation involves covalent bonds | 2               |     |     | 2     |       |      |
|          | (b) | (i)  |  | <br>Mg <sup>2+</sup>  <sup>2-</sup> (1)                                                                                                                                               |                 | 2   |     | 2     |       |      |

| Question |  |       |  | Marking details                                                                                                                                                                                                                                                                                                                                  | Marks available |          |          |           |          |          |
|----------|--|-------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |  |       |  |                                                                                                                                                                                                                                                                                                                                                  | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
|          |  | (ii)  |  | (in magnesium oxide) the ions have higher charges (1)<br><br>electrostatic attraction is greater / attraction between ions is greater / ionic bonds are stronger (1)<br><br>accept converse for both marks                                                                                                                                       | 1               | 1        |          | 2         |          |          |
|          |  | (iii) |  | 88 (3)      accept 87.5<br><br>if answer incorrect credit each correct step in method<br><br>$\frac{4.12}{58.5} = 0.0704$ (1)<br><br>$\frac{0.0704}{0.080} = 0.88$ (1)<br><br>$0.88 \times 100 = 88$ (1)<br><br>alternative method<br><br>$0.080 \times 58.5 / 4.68$ (1)<br><br>$\frac{4.12}{4.68} = 0.88$ (1)<br><br>$0.88 \times 100 = 88$ (1) |                 | 3        |          | 3         | 3        |          |
|          |  |       |  | <b>Question 5 total</b>                                                                                                                                                                                                                                                                                                                          | <b>3</b>        | <b>8</b> | <b>0</b> | <b>11</b> | <b>3</b> | <b>0</b> |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks available |     |     |       |       |      |
|----------|-----|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 6        | (a) |  |  | <p><b>Indicative content</b></p> <p><b>X</b> identified as hydrochloric acid<br/> <b>Y</b> identified as sodium chloride<br/> <b>Z</b> identified as ethanoic acid</p> <p><b>X</b> is the stronger acid – more vigorous reaction / produces hydrogen more quickly; greater temperature increase / more exothermic</p> <p><b>Y</b> does not react so cannot be an acid / must be sodium chloride<br/> <b>Y</b> must be sodium chloride – magnesium less reactive than sodium so can't displace it</p> <p><b>Z</b> is the weaker acid – less vigorous reaction / produces hydrogen less quickly; smaller temperature rise / less exothermic</p> <p>magnesium + hydrochloric acid → magnesium chloride + hydrogen<br/> <math>\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2</math></p> <p>magnesium + ethanoic acid → magnesium ethanoate + hydrogen<br/> <math>\text{Mg} + \text{CH}_3\text{COOH} \rightarrow (\text{CH}_3\text{COO})_2\text{Mg} + \text{H}_2</math></p> <p><b>5-6 marks</b><br/> All three identified; clear reasoning; good attempt at equation<br/> <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i></p> | 2               |     | 4   | 6     |       | 4    |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks available |          |          |          |          |          |
|----------|-----|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
|          |     |  |  | <p><b>3-4 marks</b><br/>At least two identified; some reasoning; reference to named salt and/or hydrogen as products<br/><i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i></p> <p><b>1-2 marks</b><br/>At least one identified; reference to gas/hydrogen as product<br/><i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i></p> <p><b>0 marks</b><br/><i>No attempt made or no response worthy of credit.</i></p> |                 |          |          |          |          |          |
|          | (b) |  |  | <p><u>iron(II)</u> ions will produce a <u>green precipitate</u> (1)</p> <p><u>iron(III)</u> ions will produce a <u>brown precipitate</u> (1)</p> <p>if no reference to precipitate award (1) for iron(II) green <b>and</b> iron(III) brown</p> <p>award (1) if correct precipitate colours given but assigned to incorrect ions</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2               |          |          | 2        |          | 2        |
|          |     |  |  | <b>Question 6 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>4</b>        | <b>0</b> | <b>4</b> | <b>8</b> | <b>0</b> | <b>6</b> |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                  | Marks available |     |     |       |       |      |
|----------|-----|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                  | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 7        | (a) |  |  | 435 (3)<br><br>if answer incorrect credit each correct step in method<br><br>$2340 - 94 = 2246$ (1)<br><br>$2246 - 941 = 1305$ (1)<br><br>$H-H = 435$ (1)<br><br>alternative method<br><br>$2340 - 94 = 2246$ (1)<br><br>$N \equiv N + 3(H-H) = 2246$ (1)<br><br>$H-H = 435$ (1) |                 |     | 3   | 3     | 3     |      |
|          | (b) |  |  | yield is lower / is too low / less ammonia is formed (1)<br><br>reaction rate is lower / too low<br>reaction is slower / is too slow (1)                                                                                                                                         | 2               |     |     | 2     |       |      |
|          | (c) |  |  | $HNO_3 + NH_3 \rightarrow NH_4NO_3$<br><br>reactants (1)<br>product (1)                                                                                                                                                                                                          |                 | 2   |     | 2     |       |      |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks available |          |          |          |          |          |
|----------|-----|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
|          | (d) |  |  | accept any of the following approaches but the second point must directly follow from the first for both marks<br><br>(leads to) high numbers of algae / microorganisms (1)<br>⇒ decomposition of algae depletes oxygen (1)<br><br>stops sunlight reaching plants below the surface (1)<br>⇒ plants unable to photosynthesise (1)<br><br>plants unable to photosynthesise (1)<br>⇒ reduction in oxygen content of water (1)<br><br>excessive plant growth stops absorption of oxygen (from air)<br>⇒ not enough oxygen for fish (1) | 2               |          |          | 2        |          |          |
|          |     |  |  | <b>Question 7 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>4</b>        | <b>2</b> | <b>3</b> | <b>9</b> | <b>3</b> | <b>0</b> |

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                  |   | Marks available |     |     |       |       |      |
|----------|-----|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                  |   | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 8        | (a) | (i)  |  | as pH increases, citric acid content decreases and sugar content increases                                                                                                                                                                                                                                                                                       | ✓ |                 |     | 2   | 2     | 2     |      |
|          |     |      |  | as acidity decreases, ascorbic acid content decreases and water content decreases                                                                                                                                                                                                                                                                                |   |                 |     |     |       |       |      |
|          |     |      |  | tomatoes are a good source of vitamin C and citric acid                                                                                                                                                                                                                                                                                                          |   |                 |     |     |       |       |      |
|          |     |      |  | citrus fruits contain ascorbic acid and a natural preservative                                                                                                                                                                                                                                                                                                   | ✓ |                 |     |     |       |       |      |
|          |     | (ii) |  | award (1) for <b>any</b> of following <ul style="list-style-type: none"> <li>the values for ascorbic acid are bigger but the unit is 1000 times smaller</li> <li>mg/100g is a smaller unit than %</li> <li>citric acid is measured in % but ascorbic acid is measured in mg/100g</li> <li>there is much more citric acid (than ascorbic acid) present</li> </ul> |   |                 |     | 1   | 1     | 1     |      |
|          | (b) | (i)  |  | $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$<br>formulae (1)<br>state symbols (1)                                                                                                                                                                                                                                   |   | 2               |     |     | 2     |       |      |

| Question |  |       |    | Marking details                                                                                                                                                                                                                                                                                     | Marks available |          |          |           |          |          |
|----------|--|-------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |  |       |    |                                                                                                                                                                                                                                                                                                     | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
|          |  | (ii)  | I  | $0.35 \times 0.021 = 0.00735$                                                                                                                                                                                                                                                                       |                 | 1        |          | 1         | 1        | 1        |
|          |  |       | II | $0.588 / 0.59$ (2)<br>if answer incorrect credit each correct step in method<br>$0.00735 \times 2 = 0.0147$ (1)<br>$\frac{0.0147}{0.025} = 0.588 / 0.59$ (1)<br>ecf possible from part (i)                                                                                                          |                 | 2        |          | 2         | 2        | 2        |
|          |  | (iii) |    | $1.2 \times 10^{22}$ (2)<br>accept $12 \times 10^{21} / 0.12 \times 10^{23}$<br>if answer incorrect credit each correct step in method but do not award any marks for multiplying mass 0.36 by $N_A$<br>$\frac{0.36}{18} = 0.020$ (1)<br>$0.020 \times 6.0 \times 10^{23} = 1.2 \times 10^{22}$ (1) |                 | 2        |          | 2         | 2        |          |
|          |  |       |    | <b>Question 8 total</b>                                                                                                                                                                                                                                                                             | <b>2</b>        | <b>5</b> | <b>3</b> | <b>10</b> | <b>8</b> | <b>3</b> |

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                 | Marks available |     |     |       |       |      |
|----------|-----|------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                 | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 9        | (a) |      |  | aluminium ions gain electrons therefore are reduced (1)<br><br>accept $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al} \quad \Rightarrow$ reduction<br><br>oxide ions lose electrons therefore are oxidised (1)<br><br>accept $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^- \quad \Rightarrow$ oxidation<br><br>neutral answer – oxidation is loss, reduction is gain | 2               |     |     | 2     |       |      |
|          | (b) | (i)  |  | $\frac{36}{100} \times 500 = 180$ tonnes of $\text{Al}_2\text{O}_3$                                                                                                                                                                                                                                                                                                             |                 | 1   |     | 1     | 1     |      |
|          |     | (ii) |  | 95.3 / 95 (3)<br><br>if answer incorrect credit each correct step in method<br><br>102 tonnes $\text{Al}_2\text{O}_3$ produces 54 tonnes of Al (1)<br><br>1 tonne $\text{Al}_2\text{O}_3$ produces $\frac{54}{102}$ tonnes of Al (1)<br><br>180 tonnes $\text{Al}_2\text{O}_3$ produces 95.3 tonnes of Al (1)<br><br>ecf possible from part (i)                                 |                 | 3   |     | 3     | 3     |      |

| Question |  |  |  | Marking details                                                                                                                                                                     | Marks available |     |     |       |       |      |
|----------|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |  |  |                                                                                                                                                                                     | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |  |  |  | alternative method<br><br>$\frac{180}{102} = 1.765 \quad (1)$<br><br>$1.765 \times 2 = 3.530 \quad (1)$<br><br>$3.530 \times 27 = 95.3 \quad (1)$<br><br>ecf possible from part (i) |                 |     |     |       |       |      |
|          |  |  |  | Question 9 total                                                                                                                                                                    | 2               | 4   | 0   | 6     | 4     | 0    |

## FOUNDATION TIER

## SUMMARY OF MARKS ALLOCATED TO ASSESSMENT OBJECTIVES

| Question     | AO1       | AO2       | AO3       | TOTAL MARK | MATHS     | PRAC      |
|--------------|-----------|-----------|-----------|------------|-----------|-----------|
| <b>1</b>     | 2         | 1         | 6         | <b>9</b>   | 0         | 8         |
| <b>2</b>     | 3         | 4         | 1         | <b>8</b>   | 4         | 2         |
| <b>3</b>     | 4         | 5         | 0         | <b>9</b>   | 5         | 4         |
| <b>4</b>     | 9         | 1         | 2         | <b>12</b>  | 0         | 0         |
| <b>5</b>     | 4         | 4         | 4         | <b>12</b>  | 4         | 0         |
| <b>6</b>     | 0         | 7         | 3         | <b>10</b>  | 2         | 0         |
| <b>7</b>     | 4         | 3         | 2         | <b>9</b>   | 0         | 3         |
| <b>8</b>     | 7         | 4         | 0         | <b>11</b>  | 3         | 4         |
| <b>TOTAL</b> | <b>33</b> | <b>29</b> | <b>18</b> | <b>80</b>  | <b>18</b> | <b>21</b> |

## HIGHER TIER

## SUMMARY OF MARKS ALLOCATED TO ASSESSMENT OBJECTIVES

| Question     | AO1       | AO2       | AO3       | TOTAL MARK | MATHS     | PRAC      |
|--------------|-----------|-----------|-----------|------------|-----------|-----------|
| <b>1</b>     | 4         | 3         | 2         | <b>9</b>   | 0         | 3         |
| <b>2</b>     | 7         | 4         | 0         | <b>11</b>  | 3         | 4         |
| <b>3</b>     | 7         | 2         | 0         | <b>9</b>   | 1         | 0         |
| <b>4</b>     | 0         | 3         | 4         | <b>7</b>   | 3         | 4         |
| <b>5</b>     | 3         | 8         | 0         | <b>11</b>  | 3         | 0         |
| <b>6</b>     | 4         | 0         | 4         | <b>8</b>   | 0         | 6         |
| <b>7</b>     | 4         | 2         | 3         | <b>9</b>   | 3         | 0         |
| <b>8</b>     | 2         | 5         | 3         | <b>10</b>  | 8         | 3         |
| <b>9</b>     | 2         | 4         | 0         | <b>6</b>   | 4         | 0         |
| <b>TOTAL</b> | <b>33</b> | <b>31</b> | <b>16</b> | <b>80</b>  | <b>25</b> | <b>20</b> |